

Graduate Algorithms

Quiz 3

August 25, 2026

Suckin at something is the first step towards being kinda good at something

Jake the Dog, Adventure Time

Instructions

- Total marks: 21, capped to 20. Time: 55 minutes.
- There are 2 problems; both are compulsory.
- Write your answers in the answer sheets provided.
- Throughout, $\lg$ denotes $\log_2$.

Question 1: Powers of Two, Written Out

6 marks

You are given the following procedure as a **black box**: you may call it, but you do not need to know how it works.

MULTIPLY(x, y)

- *Input.* The decimal representations of two non-negative integers x and y , having m and n digits respectively, where $m \leq n$.
- *Output.* The decimal representation of the product $x \cdot y$.
- *Running time.* $O(n m^{\lg 3 - 1})$. In particular, when both arguments have d digits this is $O(d^{\lg 3})$.

Assume in addition that adding two d -digit decimal numbers takes $O(d)$ time. The procedure (Algorithm 1) computes the decimal representation of 2^n . Every quantity it manipulates is the decimal representation of an integer: MULTIPLY multiplies two such representations, and $+$ denotes decimal addition of two such representations, at the costs stated above.

Algorithm 1 Decimal representation of 2^n

```
1: function POWER( $n$ )
2:   if  $n = 0$  then
3:     return 1                                ▷ decimal representation of  $2^0$ 
4:   end if
5:   if  $n = 1$  then
6:     return 2                                ▷ decimal representation of  $2^1$ 
7:   end if
8:    $h \leftarrow \lfloor n/2 \rfloor$ 
9:    $x \leftarrow \text{POWER}(h)$                       ▷ decimal representation of  $2^h$ 
10:   $y \leftarrow \text{MULTIPLY}(x, x)$                 ▷ decimal representation of  $2^{2h}$ 
11:  if  $n$  is odd then
12:     $y \leftarrow y + y$                           ▷ decimal addition
13:  end if
14:  return  $y$ 
15: end function
```

- (a) Write down a recurrence for the running time $T(n)$ of $\text{POWER}(n)$, justifying the cost of the work done outside the recursive call. **(3 marks)**
- (b) Solve your recurrence using a recursion tree. State the bound you get and say which part of the tree carries the dominant cost. **(3 marks)**

Question 2: The Second Smallest Element

15 marks

Let $A[1 \dots n]$ be an array of $n \geq 2$ *distinct* integers. Its *second smallest* element is the element of rank 2, that is, the smallest entry of A apart from the minimum.

We work in the comparison model: an algorithm may compare two entries of A and learn which is smaller, at a cost of one unit, and this is the only way it may inspect the entries. All algorithms below are deterministic.

- (a) Write pseudocode for an algorithm that returns the second smallest element of A in a single pass over the array. State the exact number of comparisons it makes in the worst case, and give its running time. Briefly justify the correctness. **(3+1+1+1 = 6 marks)**
- (b) The algorithm in (a) may not be optimal. Give pseudocode for an algorithm that returns the second smallest element using at most

$$n + \lceil \lg n \rceil - 2$$

comparisons, and show that it makes at most this many comparisons. Briefly justify the correctness. **(5+2+2 = 9 marks)**

(Try this problem after the quiz: Prove that this is best possible: every deterministic algorithm in the comparison model makes at least $n + \lceil \lg n \rceil - 2$ comparisons on some input of size n .)